ALUGOJU VARSHITH

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PROFILE SUMMARY

Computer Science and Informatics undergraduate skilled in Python, Java, C++, PyTorch, NLP, computer vision, speech processing, LLM applications, and Retrieval-Augmented Generation. Completed research at NIT Warangal on fact-verification datasets and LLM bias assessment. Built AI applications including a Manifest V3 Chrome extension, voice-cloning system, and multimodal interview-integrity platform.

SKILLS

- **Languages:** Python, Java, C++, JavaScript
- **AI/ML:** PyTorch, NLP, Computer Vision, Speech Processing, LLMs, RAG, Prompt Engineering
- **Frameworks/Tools:** MeloTTS, cVAE, Flow Models, Chrome Extension APIs, Git, GitHub, VS Code
- **Web:** HTML, CSS, REST APIs
- **Engineering:** Object-Oriented Programming, Data Structures and Algorithms, AutoCAD

WORK EXPERIENCE

Python Research Intern | National Institute of Technology (NIT) Warangal **2 Months**

- **Towards Trustworthy Large Language Models: Fact Verification Dataset Development and Bias Assessment**
- Developed a fact verification dataset to systematically evaluate the factual accuracy of responses generated by large language models including **ChatGPT, Gemini, and Claude**.
- Designed a methodology to query LLMs with verified questions and compare generated answers against ground-truth facts, classifying responses as **correct, incorrect, partially correct, biased, or neutral**.
- Applied **Retrieval-Augmented Generation (RAG)** to ground LLM responses in verified source documents, improving factual accuracy and reducing hallucinations.
- ***Tech Stack: Python, LLM APIs, RAG***

PROJECTS

- **NVYRA — AI-Powered Chrome Extension**
- Developed a **Manifest V3** Chrome extension using **JavaScript, Chrome APIs, Groq, and LLaMA 3.3 70B** to provide webpage explanations, summaries, text-selection analysis, and screen-snipping assistance.
- Implemented a floating interactive chatbot and CORS-safe background service-worker API pipeline for low-latency response streaming across active browser tabs.
- ***Tech Stack: JavaScript, Chrome Extension APIs, Groq, LLaMA 3.3 70B***
- **Voice Cloning Using Dual Attention Mechanism and Mixed Logistic Distribution**
- Developed a voice cloning framework integrating **MeloTTS, cVAE**, and flow-based voice conversion models to synthesize natural speech.
- Applied a **dual attention mechanism** and **Mixture of Logistic** distributions to enhance speech feature modeling and acoustic robustness.
- ***Tech Stack: Python, PyTorch, MeloTTS, cVAE, Flow Models***
- **Interview Integrity AI — Multimodal Candidate Monitoring System**
- Built a real-time interview monitoring system using **computer vision, speech processing, and NLP** to detect potential malpractice during remote assessments.
- Integrated face verification, gaze/head tracking, voice consistency, lip-sync verification, and AI-generated answer detection into a unified risk score.
- ***Tech Stack: Python, Computer Vision, NLP, Speech Processing, Deep Learning***

EDUCATION

B.Tech in Computer Science and Informatics — Mahatma Gandhi University, Nalgonda, Telangana **Expected 2027**

Diploma in Civil Engineering — VMR Polytechnic, Warangal **CGPA: 7.8 | 2024**

Secondary School Certificate (SSC) — Mahatma Jyothiba Phule Residential School, Hanamkonda **CGPA: 10.0 | 2021**

CERTIFICATIONS

- **Osmania University:** Hackathon Participation Certificate; **BLR AI Hackathon:** Bangalore Participation Certificate
- **QBX Hack Arena, Kanpur:** 10th Position and Participation Certificate; **NCC 'A' Certificate** (Air Wing) — A Grade
- **CRT Computer Certification** — Setwin, Hyderabad; **Drawing Lower Grade Certificate** — Directorate of Government Examinations, Telangana

ACHIEVEMENTS

- Secured **10th Position** at QBX Hack Arena, Kanpur; participated in the Osmania University Hackathon and the BLR AI Hackathon, Bangalore.
- Awarded the **NCC 'A' Certificate (Air Wing) with A Grade**.